

CYBERSECURITY ANALYSIS

Suspicious Web Threat Interactions

AWS CloudWatch WAF Traffic Analysis Report

Project Title	Cybersecurity: Suspicious Web Threat Interactions
Domain	Data Analytics
Dataset	CloudWatch_Traffic_Web_Attack.csv
Records	282 suspicious web sessions
Tools Used	Python, Scikit-learn, Plotly, NetworkX, TensorFlow
Difficulty Level	Advanced
Report Date	March 29, 2026

Prepared as part of Data Analyst Internship Project
Unified Mentor Internship Program

1. Project Overview & Objectives

This project focuses on analyzing web traffic data collected from AWS CloudWatch to detect, classify, and profile suspicious cyber activities targeting a production web server. The dataset contains 282 flagged sessions identified by AWS WAF rules, spanning an 11-hour attack window from April 25 to April 26, 2024 (UTC).

Primary Objectives

Detect Anomalies	Identify the most dangerous sessions within the suspicious traffic pool using Isolation Forest.
Profile Attackers	Build behavioral profiles for each unique source IP based on traffic patterns.
Risk Quantification	Develop a custom 0-100 risk scoring system to prioritize threat response.
Behavioral Clustering	Group attack sessions into behavioral archetypes using K-Means clustering.
Geographic Intelligence	Map attack origins globally to identify high-risk source countries.
Timeline Analysis	Analyze how the attack evolved over time and detect burst activity windows.

2. Dataset Description

The dataset was sourced from AWS CloudWatch monitoring of a production web server. All 282 records represent traffic flagged as suspicious by AWS WAF (Web Application Firewall) rules, specifically identified under the Adversary Infrastructure Interaction observation category.

Dataset Statistics

Metric	Value
Total Records	282 sessions
Total Features	16 columns
Missing Values	0 (clean dataset)
Unique Source IPs	28 distinct attackers
Source Countries	7 (US, CA, DE, AT, NL, AE, IL)
Attack Window	2024-04-25 23:00 UTC to 2024-04-26 09:50 UTC
Session Duration	600 seconds (fixed 10-min intervals)
Protocol	HTTPS only (Port 443)
Detection Type	waf_rule (100% of records)
Bytes In Range	40 bytes to 25.2 MB per session
Bytes Out Range	44 bytes to 1.56 MB per session

Key Dataset Columns

Column	Type	Description
bytes_in	int64	Bytes received by the server from source IP
bytes_out	int64	Bytes sent from the server to source IP
creation_time	datetime	Session start timestamp (UTC)
end_time	datetime	Session end timestamp (UTC)
src_ip	object	Source IP address of the attacker
src_ip_country_code	object	Country of origin (2-letter ISO code)
protocol	object	Network protocol used (HTTPS)
response.code	int64	HTTP response code (200 for all)
dst_port	int64	Destination port (443 for all)
rule_names	object	WAF rule that flagged the traffic
detection_types	object	Detection mechanism (waf_rule)

3. Exploratory Data Analysis (EDA)

EDA was performed to understand traffic distributions, country-level attack patterns, protocol usage, and byte transfer behaviors. Key insights are summarized below.

3.1 Traffic Volume Distribution

Raw byte values span an extreme range (40 bytes to 25 MB), making log-scale transformation essential for meaningful visualization. The log-transformed distribution reveals a bimodal pattern — a cluster of low-volume probe sessions and a separate cluster of high-volume data-transfer attacks.

KEY INSIGHT

Log(bytes_in) shows two distinct attack modes: lightweight reconnaissance probes (low bytes) and heavy data-exchange sessions potentially indicating exfiltration attempts.

3.2 Country-Level Attack Distribution

Country	Code	Sessions	% of Total	Avg Bytes In
United States	US	113	40.1%	~2.8 MB
Canada	CA	72	25.5%	~0.06 MB
Germany	DE	28	9.9%	~0.04 MB
Austria	AT	21	7.4%	~0.03 MB
Netherlands	NL	18	6.4%	~1.1 MB
United Arab Emirates	AE	16	5.7%	~0.03 MB
Israel	IL	14	5.0%	~0.03 MB

FINDING

US and Canada together account for 65.6% of all attack sessions — suggesting coordinated or botnet-driven activity from North American IP ranges.

3.3 Correlation Analysis

Correlation analysis revealed a very strong positive correlation ($r = 0.998$) between bytes_in and bytes_out, indicating that sessions with high inbound traffic consistently generate proportionally high server responses — a pattern consistent with web application enumeration or data extraction attacks. Response code and destination port showed zero variance (constant values of 200 and 443 respectively).

4. Feature Engineering

11 new features were engineered beyond the base columns to enrich the dataset for machine learning models and provide deeper security insights. These features capture behavioral patterns not visible in raw traffic data.

Feature	Formula / Source	Security Relevance
duration_sec	end_time - creation_time	Session length validation
throughput_in	bytes_in / duration_sec	Attack intensity (bytes/sec)
throughput_out	bytes_out / duration_sec	Response intensity
exfil_ratio	bytes_out / (bytes_in + 1)	Data exfiltration indicator
total_bytes	bytes_in + bytes_out	Total session volume
log_bytes_in	log(1 + bytes_in)	Normalized for skewed distribution
log_bytes_out	log(1 + bytes_out)	Normalized for skewed distribution
hour	creation_time.hour	Time-of-day attack pattern
is_nighttime	1 if hour < 6 or hour > 21	Stealth attack indicator
country_encoded	LabelEncoder(src_ip_country_code)	ML-compatible country feature
ip_frequency	count(src_ip) across dataset	Persistent attacker indicator

KEY FEATURE
— exfil_ratio

The exfiltration ratio (bytes_out / bytes_in) is the most security-critical engineered feature. A ratio > 2.0 means the server sent back more than twice the data it received — potentially indicating successful data extraction by the attacker.

5. Machine Learning Models & Results

5.1 Random Forest Classifier

A Random Forest Classifier with 200 estimators was trained on 9 engineered features. The model achieved 100% accuracy on the test set. The primary value is the feature importance ranking, which reveals which traffic characteristics are most predictive.

Rank	Feature	Importance	Interpretation
1	log_bytes_in	High	Volume of inbound data is the strongest signal
2	ip_frequency	High	Repeat IPs are the most consistent attackers
3	exfil_ratio	Medium	Data return ratio differentiates attack types
4	throughput_in	Medium	Attack speed distinguishes burst vs slow attacks
5	country_encoded	Medium	Geography contributes moderate predictive power
6	log_bytes_out	Low-Med	Server response volume adds supporting signal
7	hour	Low	Time of attack has minor but measurable effect
8	is_nighttime	Low	Night attacks slightly more anomalous
9	throughput_out	Low	Response speed provides minimal extra signal

5.2 K-Means Clustering — Attacker Behavioral Profiling

K-Means clustering (k=4) was applied to group the 282 attack sessions into behavioral archetypes without using any labels. The elbow method confirmed 4 as the optimal cluster count.

Cluster	Profile Name	Characteristics	Risk
0	Reconnaissance Probe	Low bytes, low frequency, short bursts	LOW
1	Persistent Attacker	High frequency, moderate bytes, multi-hour	HIGH
2	Data Exfiltrator	High exfil ratio, high bytes_out	CRITICAL
3	Burst Attack	Very high bytes_in, few sessions, single IP	HIGH

5.3 Anomaly Detection — Isolation Forest

Isolation Forest was applied with a contamination rate of 10% to identify the most statistically extreme sessions within the already-suspicious dataset. It identifies sessions that are anomalous even within a pool of malicious traffic — the most dangerous outliers.

RESULT	Isolation Forest flagged approximately 28 sessions (~10%) as critical anomalies. These sessions showed extreme bytes_in values, unusually high exfiltration ratios, and came predominantly from US and NL source IPs.
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6. Custom Risk Scoring System

A quantitative risk scoring system (0-100 scale) was developed to prioritize threat response — mimicking real-world Security Operations Center (SOC) workflows. Each session is assigned a composite score based on four weighted components:

Component	Weight	Feature Used	Rationale
Data Volume	30%	log_bytes_in	Larger data transfers indicate more severe attacks
Exfil Potential	25%	exfil_ratio	High server response may indicate data leak
IP Persistence	25%	ip_frequency	Repeat attackers pose greater sustained threat
Attack Speed	20%	throughput_in	Faster attacks suggest automated/botnet tools

Risk Tier Classification

Tier	Score	Action Required	Sessions
CRITICAL	75-100	Immediate block + incident investigation	~12
HIGH	50-74	Block IP + alert security team	~28
MEDIUM	25-49	Rate-limit + monitor closely	~220
LOW	0-24	Log and periodic review	~22

7. Key Findings & Recommendations

7.1 Key Security Findings

F1	North American Dominance US (40.1%) and Canada (25.5%) together account for 65.6% of all attack sessions. This concentration suggests possible botnet infrastructure or cloud-hosted attack tools operating from North American IP ranges.
F2	HTTPS Evasion Strategy 100% of attacks use HTTPS on port 443 — the standard web traffic port. This deliberately blends attack traffic with legitimate traffic, evading basic port-based firewall rules and making detection dependent on behavioral analysis rather than network layer filtering.
F3	High Correlation Between In/Out Bytes bytes_in and bytes_out show near-perfect correlation (r=0.998), indicating the server is actively responding to attack requests. WAF is flagging but NOT blocking traffic — a configuration that should be reviewed immediately.
F4	Fixed Session Windows All sessions are exactly 600 seconds (10 minutes), suggesting automated tools with fixed timing intervals — a strong indicator of scripted or botnet-based attacks.
F5	28 Unique Source IPs Only 28 distinct source IPs generated all 282 sessions. Several IPs appear 15-20+ times, indicating persistent, repeat attackers — primary targets for immediate blocking.
F6	Attack Timeline Burst at 08:00-10:00 UTC Timeline analysis revealed a significant spike during the 08:00-10:00 UTC window on April 26 — suggesting a coordinated second wave of attacks during business hours.

7.2 Recommendations

Immediate Actions

- Block all 28 identified source IPs at the WAF/firewall level immediately.
- Investigate the ~28 Isolation Forest anomaly sessions for indicators of compromise.
- Review WAF configuration — traffic is flagged but not blocked (HTTP 200 responses).

Short-Term Actions

- Implement rate-limiting for IPs with frequency > 10 sessions per hour.
- Add geo-blocking rules for countries not in expected business regions.
- Deploy real-time alerting on exfil_ratio metric exceeding 2.0 threshold.

Long-Term Actions

- Integrate this risk scoring model into the live WAF pipeline for automated response.

- Expand dataset collection to include legitimate traffic for comparative baseline analysis.
- Schedule monthly re-training of clustering model to adapt to evolving attacker patterns.